Wireshark Network Diagnostics for Mac Users

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COMM-2216: Business Communications 2

Manual Instruction

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Introduction

Wireshark is a network protocol analyzer, a software tool that captures and displays packets of data travelling through a network interface. You can use Wireshark to examine how your computer communicates with other devices and identify where a connection may be failing. For example, if your Mac cannot use a network printer while other employees can print normally, you can inspect the traffic between your Mac and the printer to help locate the problem.

This manual is intended for Mac users who are IT students or entry-level IT employees. You should know how to open applications, connect to Wi-Fi or Ethernet, and enter commands in macOS Terminal. Previous experience with Wireshark is not required; the screenshots and glossary support readers who are learning to interpret network traffic.

The five tasks cover problems with a network printer, Internet access, a company shared drive, a specific website, and a slow Internet connection. Depending on the task, you will examine address resolution, connectivity tests, name lookups, connection establishment, or response times. These checks help you collect evidence to report to your IT support team; they do not automatically repair the connection.

For the setup used in this manual, prepare a Mac with an Apple M1 chip or later, Wireshark 4.6.8, a compatible version of macOS, Terminal, and a web browser.

Choose the task that matches your problem and follow its checks in order. Capture only traffic you are authorized to inspect, and protect capture files because they may contain sensitive information. If you cannot capture or locate packets, consult the troubleshooting guide. For additional help, consult the online Wireshark User's Guide at www.wireshark.org/docs/wsug_html/.

Sets of Instructions

Task 1: Diagnosing the connection between your computer and a network printer

Overview

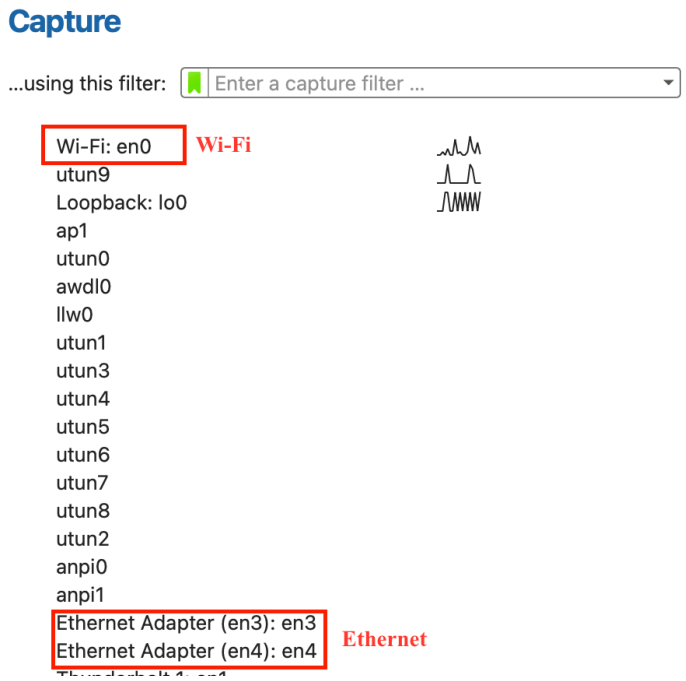
If your computer cannot connect to a network printer while other employees can use it normally, you can use Wireshark to determine where the connection fails. Please open Wireshark and check the following three areas where a connection problem may occur:

1. ARP
2. ICMP
3. TCP

Step 1: Check ARP

1. Click **Wi-Fi: en0** in the Capture section if your computer is using a wireless connection, or click **an Ethernet adapter** if your computer is using a wired connection (see Figure 1).

Figure 1 - Selecting a network interface in Wireshark



2. Click the blue Shark-Fin Icon to start capturing packets.

3. Type `arp` in the Display Filter Box and press **Enter**.

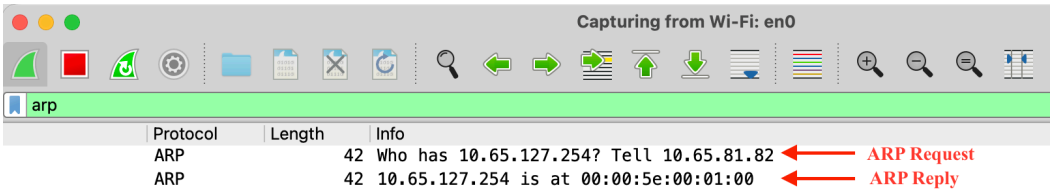
4. Type `ping <printer-IP>` in macOS Terminal and press **Enter**.

Explanation: Replace <printer-IP> with the actual IP address of the network printer. You can type `lpstat -v` in macOS Terminal to view the addresses of configured network printers.

Example: If the printer address is ipps:// 10.65.127.254:9164/printers/DTC_Payprint, then the printer's IP is 10.65.127.254.

5. Locate the ARP Request and ARP Reply for the printer's IP address in Wireshark, (see Figure 2, suppose the printer's IP address is 10.65.127.254).

Figure 2 - ARP Request and ARP Reply in Wireshark



Tip: If there are too many ARP Requests and ARP Replies to locate the printer easily, type `arp.src.proto_ipv4 == <printer-IP> || arp.dst.proto_ipv4 == <printer-IP>` in the Display Filter Box and press **Enter** to narrow the results.

Note: If an ARP Reply is received, then continue to Step 2. Otherwise, stop here and investigate the ARP problem further.

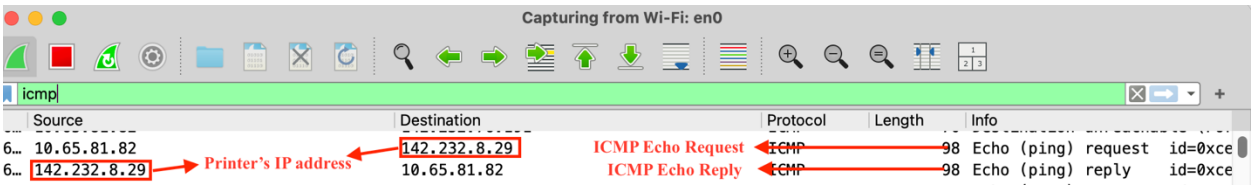
Step 2: Check ICMP

1. Keep the macOS Terminal open from Step 1.

2. Type `icmp` in the Display Filter Box and press **Enter**.

3. Locate the ICMP Echo Request and Echo Reply for the printer's IP address in Wireshark, (see Figure 3, suppose the printer's IP address is 142.232.8.29).

Figure 3 – ICMP Echo Request and ICMP Echo Reply in Wireshark



Tip: If there are too many ICMP Echo Requests and ICMP Echo Replies to locate the printer easily, type `icmp && ip.addr == <printer-IP>` in the Display Filter Box and press **Enter** to narrow the results.

Note: If an Echo Reply is received, then continue to Step 3. Otherwise, stop here and investigate the IP connectivity problem further.

Step 3: Check TCP

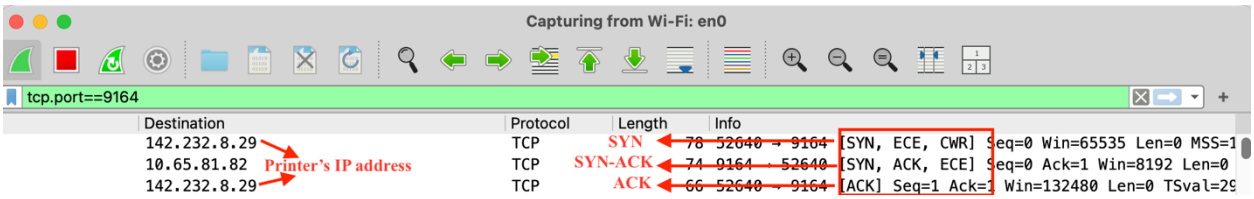
1. Keep the macOS Terminal open from Step 2.
2. Type `tcp.port == <port-number>` in the Display Filter Box and press **Enter**.

Explanation: Replace <port-number> with the actual port number used by the network printer. You can type `lpstat -v` in macOS terminal to find the printer connection information and its port number.

Example: If the printer address is ipps://142.232.8.29:9164/printers/DTC_Payprint, the port number is 9164.

3. Print a document using the related network printer.
4. Locate the TCP three-way handshake: SYN → SYN-ACK → ACK in Wireshark, (see Figure 4).

Figure 4 – TCP Three-Way Handshake in Wireshark



Note: If Wireshark displays the TCP three-way handshake, then the TCP connection to the printing service is working. Otherwise, stop here and investigate the TCP connection problem further.

🙂 Great! You have completed all the necessary checks. You can now use the results to identify the connection problem and report it to your IT support team for further assistance.

Task 2: Diagnosing a computer that cannot access the Internet

Overview

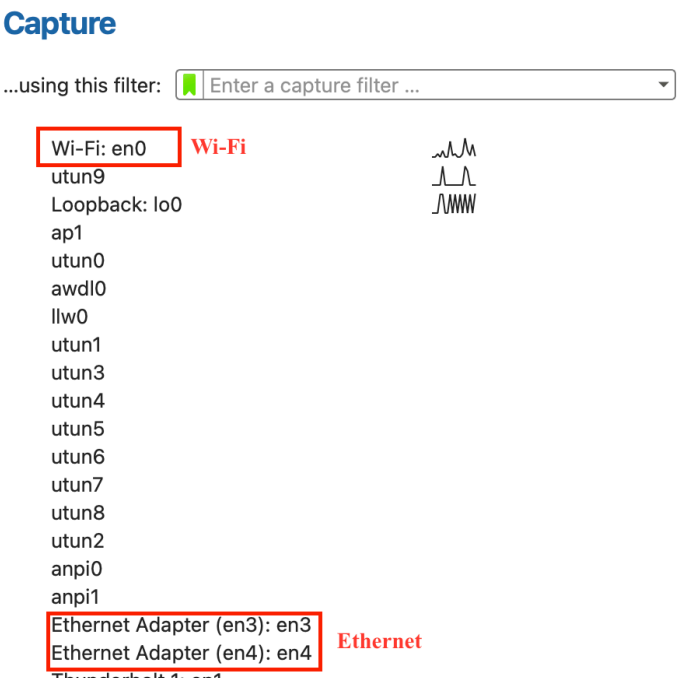
If your computer cannot access the Internet while other employees can access it normally, you can use Wireshark to determine where the connection fails. Please open Wireshark and check the following three areas where a connection problem may occur:

1. ARP
2. ICMP
3. TCP

Step 1: Check ARP

1. Click **Wi-Fi: en0** in the Capture section if your computer is using a wireless connection, or click **an Ethernet adapter** if your computer is using a wired connection (see Figure 5).

Figure 5 - Selecting a network interface in Wireshark



2. Click the blue Shark-Fin Icon to start capturing packets.

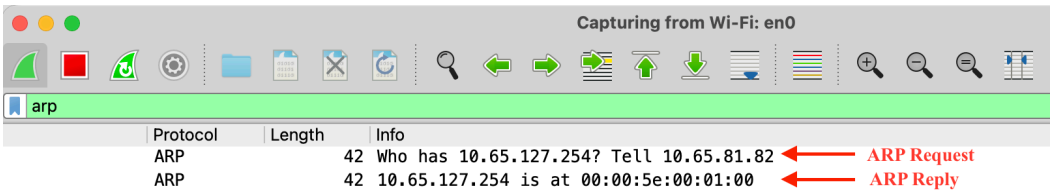
3. Type `arp` in the Display Filter Box and press Enter.

4. Type `ping <default-gateway-IP>` in macOS Terminal and press **Enter**.

Explanation: <default-gateway-IP> is the IP address of the router your computer uses to reach other networks. You can type `route -n get default | grep gateway` in macOS Terminal to find it.

5. Locate the ARP Request and ARP Reply for the default gateway's IP address in Wireshark, (see Figure 6, suppose the default gateway's IP address is 10.65.127.254).

Figure 6 - ARP Request and ARP Reply in Wireshark



Tip: If there are too many ARP Requests and ARP Replies to locate the default gateway easily, type `arp.src.proto_ipv4 == < default-gateway-IP > || arp.dst.proto_ipv4 == < default-gateway-IP >` in the Display Filter Box and press **Enter** to narrow the results.

Note: If an ARP Reply is received, then continue to Step 2. Otherwise, stop here and investigate the ARP problem further.

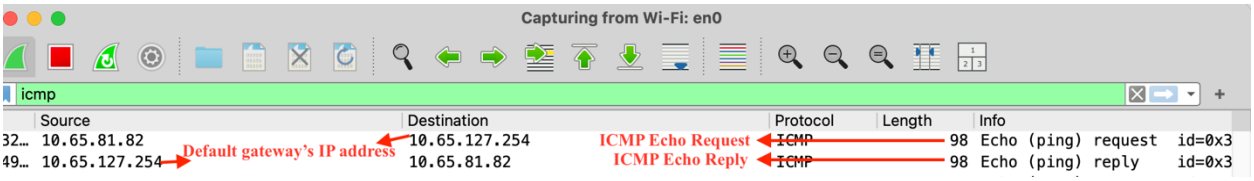
Step 2: Check ICMP

1. Keep the macOS Terminal open from Step 1.

2. Type `icmp` in the Display Filter Box and press **Enter**.

3. Locate the ICMP Echo Request and Echo Reply for the default gateway's IP address in Wireshark, (see Figure 7, suppose the default gateway's IP address is 10.65.127.254).

Figure 7 - ICMP Echo Request and ICMP Echo Reply in Wireshark



Tip: If there are too many ICMP Echo Requests and ICMP Echo Replies to locate the default gateway easily, type `icmp && ip.addr == < default-gateway-IP >` in the Display Filter Box and press **Enter** to narrow the results.

Note: If an Echo Reply is received, then continue to Step 3. Otherwise, stop here and investigate the IP connectivity problem further.

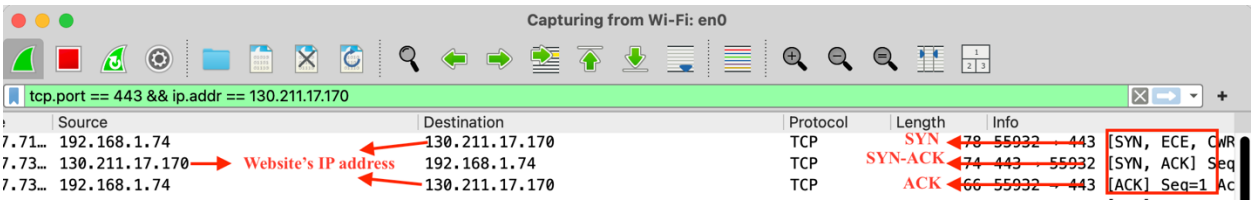
Step 3: Check TCP

1. Keep the macOS Terminal open from Step 2.
2. Type `tcp.port == 443` in the Display Filter Box and press **Enter**.

Explanation: Port 443 is commonly used for secure website connections (HTTPS).

3. Open a web browser and visit a website.
4. Locate the TCP three-way handshake: SYN → SYN-ACK → ACK in Wireshark**,** (see Figure 8, suppose the visited website's IP address is 130.211.17.170).

Figure 8 - TCP Three-Way Handshake in Wireshark



Explanation: You can type `nslookup <website-name>` in macOS Terminal to find the visited website's IP address.

Tip: If there are too many TCP packets, you can type type `tcp.port == 443 && ip.addr == <website-IP>` in the Display Filter Box and press **Enter** to narrow the results. Replace <website-IP> with the actual visited website's IP address.

Note: If Wireshark displays the TCP three-way handshake, then the TCP connection is working. Otherwise, stop here and investigate the TCP connection problem further.

🙂 Great! You have completed all the necessary checks. You can now use the results to identify the connection problem and report it to your IT support team for further assistance.

Task 3: Diagnosing a computer that cannot access a Company Shared Drive

Overview

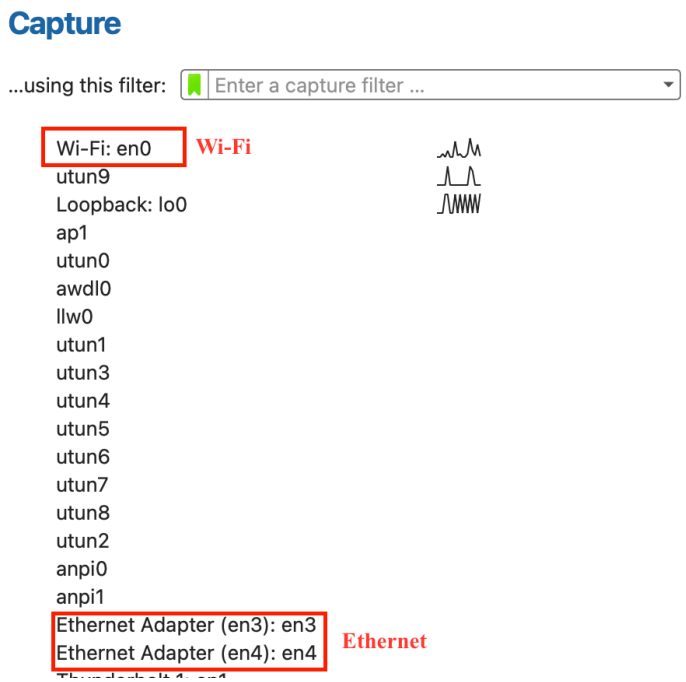
If your computer cannot access the company's shared drive while other employees can access it normally, you can use Wireshark to determine where the connection fails. Please open Wireshark and check the following two areas where a connection problem may occur:

1. ARP
2. ICMP

Step 1: Check ARP

1. Click **Wi-Fi: en0** in the Capture section if your computer is using a wireless connection, or click **an Ethernet adapter** if your computer is using a wired connection (see Figure 9).

Figure 9 - Selecting a network interface in Wireshark



2. Click the blue Shark-Fin Icon  to start capturing packets.

3. Type `arp` in the Display Filter Box and press **Enter**.

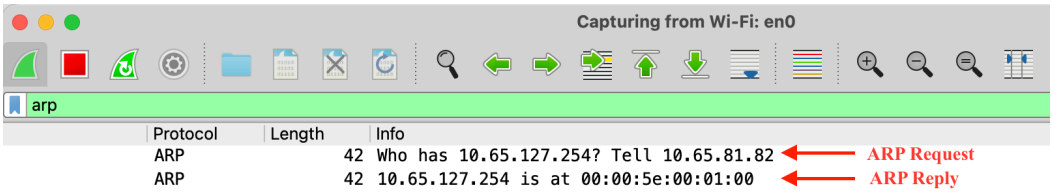
4. Type `ping <shared-drive-IP>` in macOS Terminal and press **Enter**.

Explanation: <shared-drive-IP> is the IP address of the server that hosts the company's shared drive. You can type `nslookup <server-name>` in macOS Terminal to find its IP address.

Example: If the company's shared drive address is smb://fileserver.company.com/shared, then the <server-name> is fileserver.company.com. Type `nslookup fileserver.company.com` in macOS Terminal to find its IP address. If the result is 10.65.127.254, use 10.65.127.254 as the <shared-drive-IP>.

5. Locate the ARP Request and ARP Reply for the shared drive's IP address in Wireshark, (see Figure 10, suppose the shared drive's IP address is 10.65.127.254).

Figure 10 - ARP Request and ARP Reply in Wireshark



Tip: If there are too many ARP Requests and ARP Replies, type `arp.src.proto_ipv4 == <shared-drive-IP> || arp.dst.proto_ipv4 == <shared-drive-IP>` in the Display Filter Box and press **Enter**.

Note: If an ARP Reply is received, then continue to Step 2. Otherwise, stop here and investigate the ARP problem further.

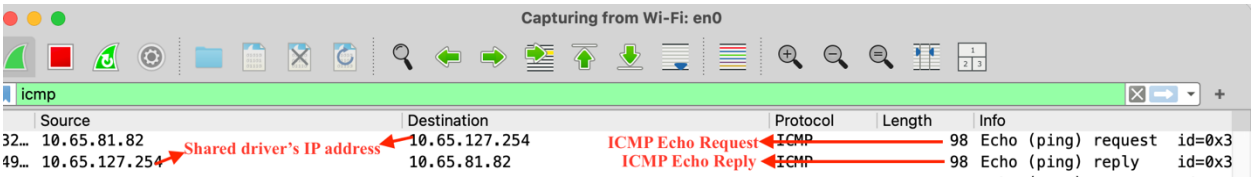
Step 2: Check ICMP

1. Keep the macOS Terminal open from Step 1.

2. Type `icmp` in the Display Filter Box and press **Enter**.

3. Locate the ICMP Echo Request and Echo Reply for the shared drive's IP address in Wireshark, (see Figure 11, suppose the shared drive's IP address is 10.65.127.254).

Figure 11 - ICMP Echo Request and ICMP Echo Reply in Wireshark



Tip: If there are too many ICMP Echo Requests and ICMP Echo Replies to locate the default gateway easily, type `icmp && ip.addr == < shared-drive-IP >` in the Display Filter Box and press **Enter** to narrow the results.

Note: If an Echo Reply is received, then the IP connection is working. Otherwise, stop here and investigate the IP connection problem further.

🙂 Great! You have completed all the necessary checks. You can now use the results to identify the connection problem and report it to your IT support team for further assistance.

Task 4: Diagnose a Computer That Cannot Access a Specific Website

Overview

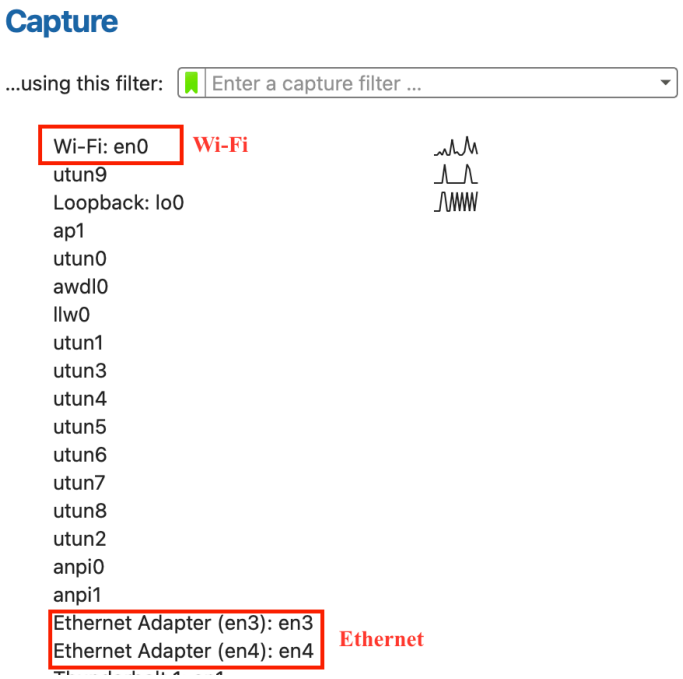
If your computer cannot access one specific website while other websites work normally, please open Wireshark and check the following three areas where a problem may occur:

1. DNS
2. TCP
3. TLS

Step 1: Check DNS

1. Click **Wi-Fi: en0** in the Capture section if your computer is using a wireless connection, or click **an Ethernet adapter** if your computer is using a wired connection (see Figure 12).

Figure 12 - Selecting a network interface in Wireshark



2. Click the blue Shark-Fin Icon  to start capturing packets.

3. Type `dns` in the Display Filter Box and press **Enter**.

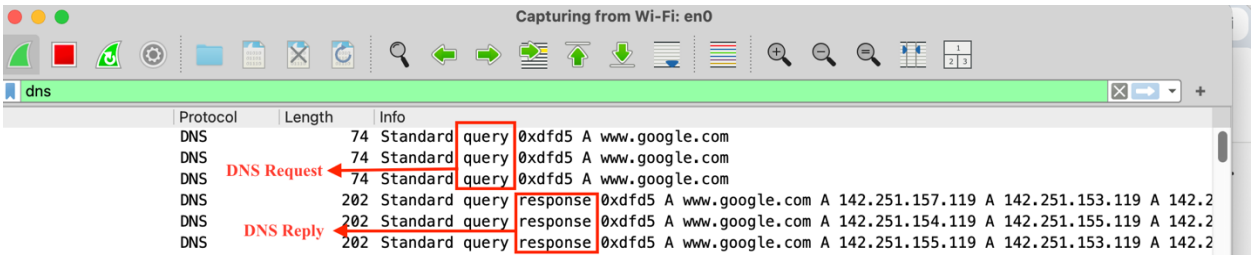
4. Type `nslookup <specific-website-name>` in macOS Terminal and press **Enter**.

Explanation: Replace < specific-website-name > with the specific website that your computer can't get access to.

Example: If the specific website that your computer can't get access to is www.google.com, then type `nslookup www.google.com` in macOS Terminal.

5. Locate the DNS Request and DNS Reply in Wireshark, (see Figure 13).

Figure 13 - DNS Request and DNS Reply in Wireshark



Tip: If there are too many DNS packets, click the Stop Capturing Packets button  , and then click the Start Capturing Packets button  to start a new capture.

Note: If the DNS Request and DNS Reply are received, then continue to Step 2. Otherwise, stop here and investigate the DNS problem further.

Step 2: Check TCP

1. Keep the macOS Terminal open from Step 1.

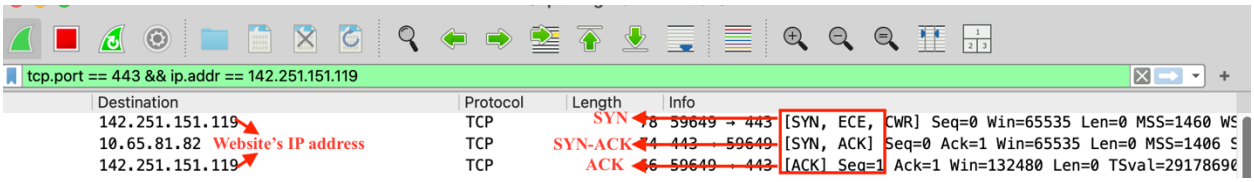
2. Type `tcp.port == 443` in the Display Filter Box and press **Enter**.

Explanation: Port 443 is commonly used for secure website connections (HTTPS).

3. Open a web browser and visit the specific website.

4. Locate the TCP three-way handshake: SYN → SYN-ACK → ACK in Wireshark**,** (see Figure 14, suppose the specific website's IP address is 142.251.151.119).

Figure 14 - TCP Three-Way Handshake in Wireshark



Explanation: You can type `nslookup <website-name>` in macOS Terminal to find the specific website's IP address.

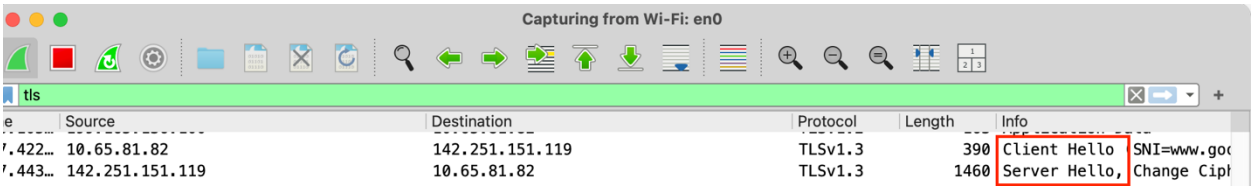
Tip: If there are too many TCP packets, you can type `nslookup <website-name>` in macOS Terminal to find the website's IP address. Then type `tcp.port == 443 && ip.addr == <website-IP>` in the Display Filter Box and press **Enter** to narrow the results. Replace <website-IP> with the actual specific website's IP address.

Note: If Wireshark displays the TCP three-way handshake, then continue to Step 3. Otherwise, investigate the TCP connection problem further.

Step 3: Check TLS

1. Keep the macOS Terminal open from Step 2.
2. Type `tls` in the Display Filter Box and press **Enter**.
3. Reload the specific website in your web browser.
4. Locate the Client Hello and corresponding Server Hello packets in Wireshark (see Figure 15).

Figure 15 - Client Hello and Server Hello packets in Wireshark



Note: If Wireshark displays the Client Hello and corresponding Server Hello packets, then the TLS connection to the specific website is working. Otherwise, stop here and investigate the TLS connection problem further.

🙂 Great! You have completed all the necessary checks. You can now use the results to identify the connection problem and report it to your IT support team for further assistance.

Task 5: Diagnose a Computer with a Slow Internet Connection

Overview

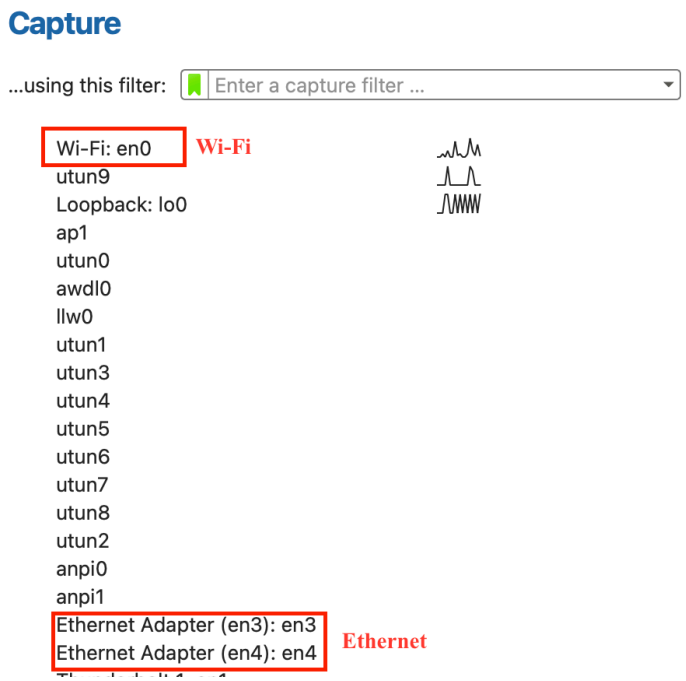
If the Internet works on your computer but is extremely slow, please open Wireshark and check the following two areas where a problem may occur:

1. ICMP
2. DNS

Step 1: Check ICMP

1. Click **Wi-Fi: en0** in the Capture section if your computer is using a wireless connection, or click **an Ethernet adapter** if your computer is using a wired connection (see Figure 16).

Figure 16 - Selecting a network interface in Wireshark



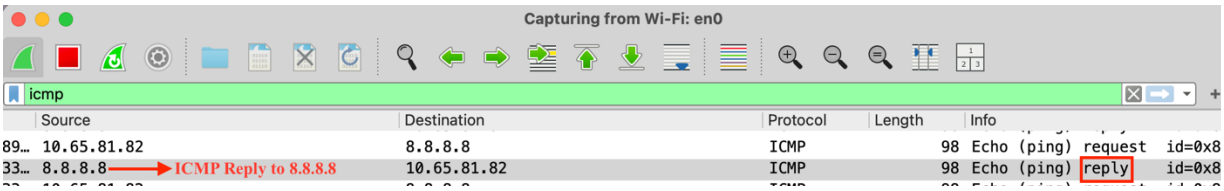
2. Click the blue Shark-Fin Icon to start capturing packets.

3. Type `icmp` in the Display Filter Box and press **Enter**.

4. Type `ping 8.8.8.8` in macOS Terminal and press **Enter**.

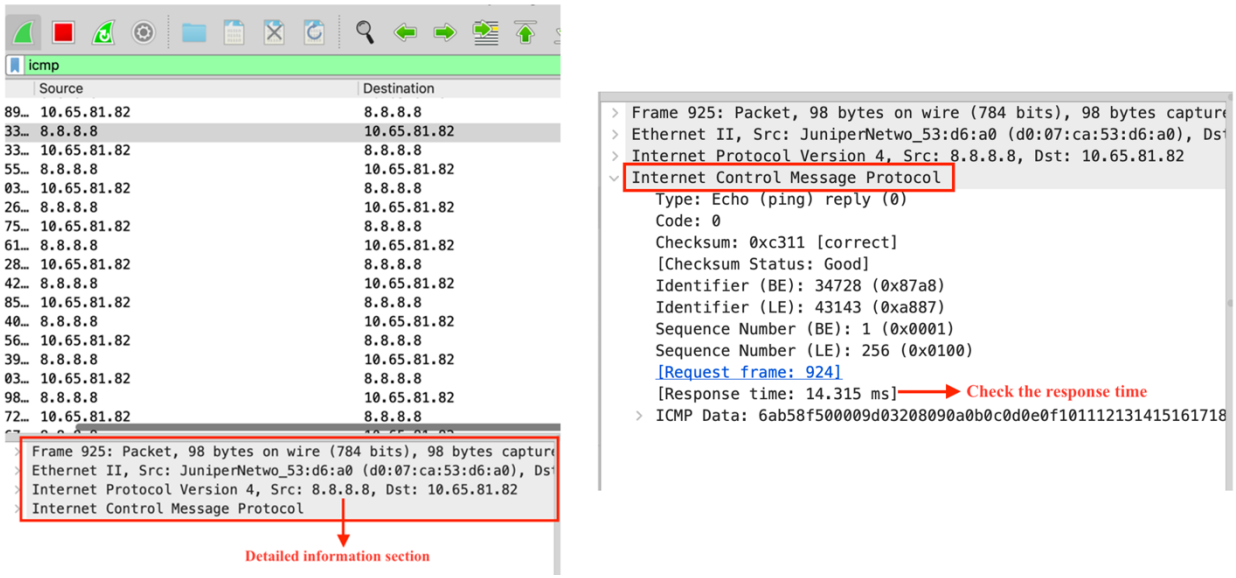
5. Locate a related ICMP Reply in Wireshark, (see Figure 17).

Figure 17 – Related ICMP Reply in Wireshark



6. Click **Internet Control Message Protocol** and check the response time in the detailed information section, (see Figure 18).

Figure 18 – Check the response time



Note: If the response time is less than 100 ms, then continue to Step 2. Otherwise, investigate the problem of ICMP delay further.

Step 2: Check DNS

1. Keep the macOS Terminal open from Step 1.

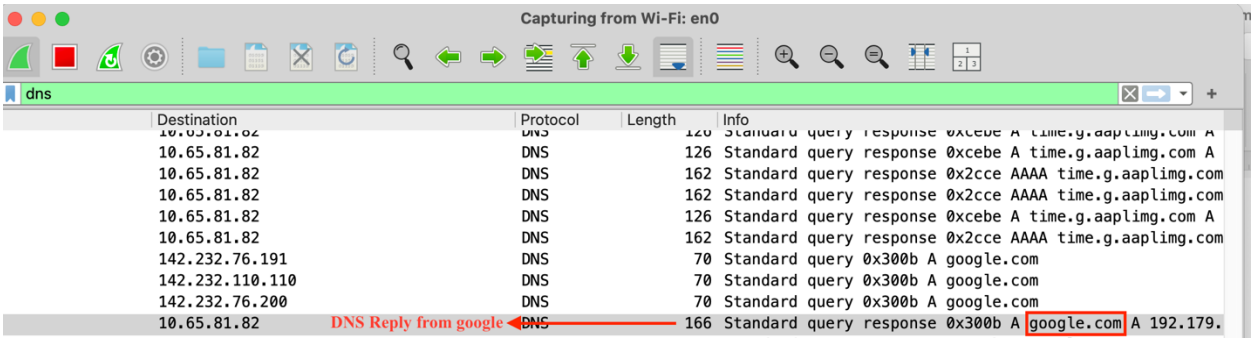
2. Type `dns` in the Display Filter Box and press **Enter**.

3. Type `nslookup <specific-website-name>` in macOS Terminal and press **Enter**.

Explanation: Replace < specific-website-name > with the actual website name. For example, it can be www.google.com.

4. Locate the DNS Reply from google in Wireshark, (see Figure 19, suppose the website being looked up is www.google.com).

Figure 19 – DNS Reply in Wireshark



5. Click **Domain Name System (response)** and check the response time in the detailed information section, (Figure 20 shows the location of the detailed information section, and Figure 21 shows how to check the response time).

Figure 20 – The detailed information section

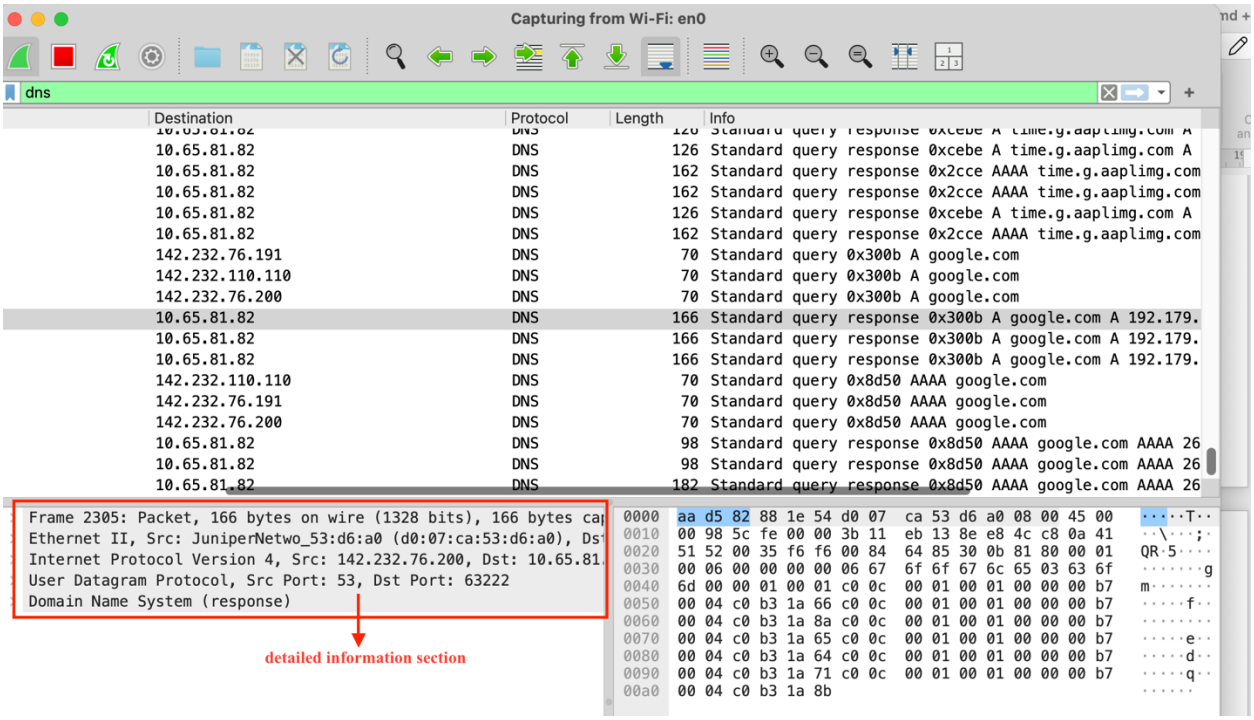
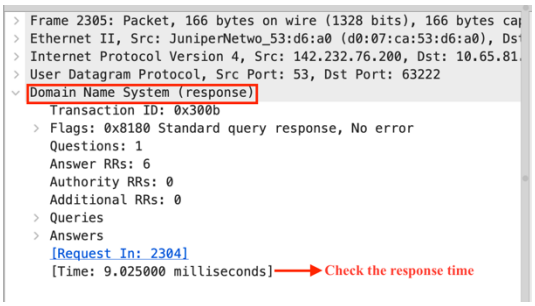


Figure 21 – Check the response time



Note: If the response time is less than 100 ms, then there is no problem in DNS delay. Otherwise, investigate the problem of DNS delay further.

🙂 Great! You have completed all the necessary checks. You can now use the results to identify the connection problem and report it to your IT support team for further assistance.

Glossary

The following table lists the terms and their definitions used in this instruction manual.

Table 1: List of Terms

Term	Definition
ARP (Address Resolution Protocol)	A network protocol that matches an IPv4 address to a MAC address on the local network.
Default gateway	A router that forwards traffic from your local network to other networks.
Display filter	A Wireshark expression that shows matching captured packets without deleting the others.
DNS (Domain Name System)	A naming system that translates domain names into IP addresses.
ICMP (Internet Control Message Protocol)	A network protocol that carries diagnostic messages, including ping requests and replies.
IP address	A numerical address that identifies a network interface for communication across IP networks.
MAC address	A link-layer address that identifies a network interface on a local network.
Network interface	A connection point, such as Wi-Fi or Ethernet, through which a computer sends and receives data.
Packet	A unit of network data containing control information and, often, a data payload.
Port number	A numerical identifier that directs TCP or UDP traffic to a particular service or application.
TCP three-way handshake	A TCP connection-opening exchange consisting of SYN, SYN-ACK, and ACK packets.
TLS (Transport Layer Security)	A security protocol that protects data in transit; HTTPS uses TLS to secure web communication.

Troubleshooting lists

Capture Permission Denied

Wireshark cannot capture packets because the user does not have permission to access the network interface. One possible cause and solution is as follows:

ChmodBPF Is Not Installed

> ***Check:*** Does Wireshark display "Permission denied" and ask you to install ChmodBPF?
>
> ***Fix:*** Close the error message, click **"Install ChmodBPF"** on the Wireshark main screen, and complete the installation.

No Packets Appear

Wireshark does not display any packets because the user may have selected an inactive network interface. One possible cause and solution is as follows:

Incorrect Network Interface

> ***Check:*** On the Wireshark main screen at the Capture section, check whether the selected interface shows network activity.
>
> ***Fix:*** Stop the capture and click the active interface, such as Wi-Fi: en0 or an Ethernet Adapter, then start capturing again.

Too Many Packets Appear

Wireshark may capture a large number of packets, making it difficult to find the traffic you need. One possible cause and solution is as follows:

No Display Filter Applied

> ***Check:*** Check whether the Display Filter Box at the top of Wireshark is empty.
>
> ***Fix:*** Type a display filter such as `dns`, `arp`, `icmp`, or `tcp` in the Display Filter Box to show only the traffic you need.